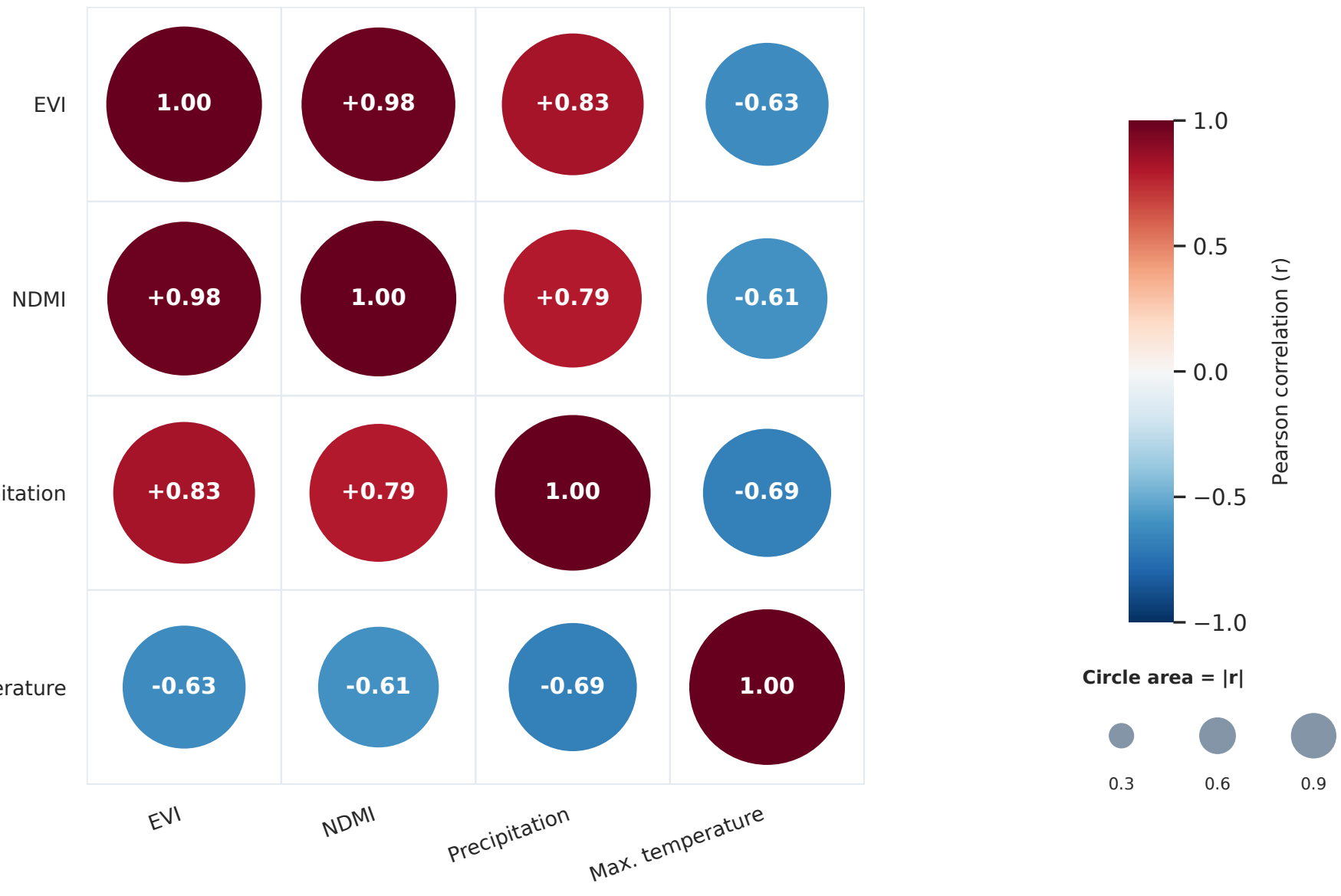
VEGETATION–CLIMATE CORRELATION STRUCTURE

Environmental Data Cube for Oglala Lakota

Ate Makoce | May–September 2000–2023 | 24 annual growing seasons

Full Pearson correlation matrix



RESULTS INTERPRETATION

EVI and NDMI exhibit the strongest positive association (r = +0.98). Precipitation is positively associated with both vegetation indices, while maximum temperature is negatively associated with vegetation and precipitation.

Interpretive limitation: These Pearson correlations describe linear associations, not causation. Annual observations may be temporally dependent, and vegetation and climate data may represent different spatial footprints.

Source: Annual EVI and NDMI validation series; gridMET precipitation and maximum temperature. May–September, 2000–2023.